Symplectic Topology

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Based on the book "Introduction to Symplectic Topology" *by Dusa McDuff & Dietmar Salamon*

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I. Foundations

1 Symplectic Linear Algebra

1.1 Symplectic spaces

Definition 1.1.1

Let V be a finite-dimensional vector space. The **kernel** of a 2-form $\omega \in \Lambda^2(V^*)$ is the subspace

$$\ker \omega = \{v \in V \mid \omega(v, \bullet) = 0\}.$$

A 2-form is **nondegenerate** if its kernel is zero.
A nondegenerate 2-form is called a **symplectic form** and the pair (V, ω) a **symplectic vector space**.

Example 1.1.2

The archetypal symplectic vector space is $\mathbb{R}^{2n}$ equipped with the 2-form

$$\omega_0 = \sum_{i=1}^{n} dx_i \wedge dy_i$$

where $x_1, \dots, x_n, y_1, \dots, y_n$ forms the standard basis and dx_i, dy_i form its dual basis.

It is clear that a 2-form ω is nondegenerate iff its representation matrix relative to any basis is non-singular. The representation matrix relative to $B = (v_1, \dots, v_n)$ is $\omega_{ij} = \omega(v_i, v_j)$. Equivalently, if the induced map $\widehat{\omega}\colon V \to V^*$ given by $\widehat{\omega}(v)(u) = \omega(v, u)$ is injective, equivalently an isomorphism.

Definition 1.1.3

Let (V, ω_V) and (W, ω_W) be two symplectic spaces. A **symplectomorphism** is a linear isomorphism $\varphi\colon V \to W$ such that the pullback of ω_W is ω_V: $\omega_V = \varphi^*\omega_W$. Recall that the pullback is defined by

$$(\varphi^*\omega_W)(v, u) = \omega_W(\varphi v, \varphi u).$$

It is clear that the composition and inverse of symplectomorphisms are symplectomorphisms, so the category of symplectic spaces with symplectomorphisms forms a groupoid.
The collection of symplectomorphisms $V \to V$ is a group denoted by $\mathrm{Sp}(V, \omega)$.

Example 1.1.4

In the case of the standard symplectic structure on $\mathbb{R}^{2n}$, we use the notation $\mathrm{Sp}(2n) = \mathrm{Sp}(\mathbb{R}^{2n}, \omega_0)$. Notice that the representation of ω_0 relative to the standard basis is

$$J_0 = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}.$$

Thus $\mathrm{Sp}(2n)$ can be viewed as the subgroup of $\mathrm{GL}_{2n}(\mathbb{R})$ defined by

$$\mathrm{Sp}(2n) = \left\{A \in \mathrm{GL}_{2n}(\mathbb{R}) \;\middle|\; A^\top J_0 A = J_0\right\}.$$

Proposition 1.1.5

$\mathrm{Sp}(2n)$ is a Lie group with Lie algebra

$$\mathfrak{sp}(2n) = \{-J_0 S \mid S \in \mathrm{Mat}_{2n} \text{ symmetric}\}.$$

Proof

Let $\mathfrak{so}(2n) \leq \mathrm{Mat}_{2n}$ be the linear subspace of skew-symmetric matrices, and define the map

$$f\colon \mathrm{Mat}_{2n} \to \mathfrak{so}(2n), \qquad f(A) = A^\top J_0 A.$$

Then $\mathrm{Sp}(2n) = f^{-1}J_0$, so to show that $\mathrm{Sp}(2n)$ is a Lie subalgebra of $\mathrm{GL}_{2n}(\mathbb{R})$, it is sufficient to show that J_0 is a regular value of f.

Note that f's differential at $A \in \mathrm{Mat}_{2n}$ is

$$df_A(X) = X^\top J_0 A + A^\top J_0 X.$$

If $f(A) = J_0$ and $B \in \mathfrak{so}(2n)$ (which is equal to $\mathfrak{so}(2n)$'s tangent space), then $X = -\frac{1}{2}AJ_0B$ is a preimage of B. Thus df_A is surjective so that J_0 is a regular value as desired.

Now,

$$\mathfrak{sp}(2n) = T_I \mathrm{Sp}(2n) = \ker df_I = \left\{X \in \mathrm{Mat}_{2n} \;\middle|\; X^\top J_0 + J_0 X = 0\right\}.$$

Note that if $X \in \mathfrak{sp}(2n)$ then $X = -J_0 S$ where $S = J_0 X$, and this is symmetric:

$$S^\top = -X^\top J_0 = J_0 X = S.$$

Conversely for S symmetric, $-J_0 S \in \mathfrak{sp}(2n)$. ∎

Definition 1.1.6

Let (V, ω) be a symplectic space and $W \leq V$ a subspace. Its **symplectic complement** is the subspace

$$W^\omega = \{v \in V \mid \omega(v, w) = 0 \text{ for all } w \in W\}.$$

Note that unlike the case for inner products, W^ω and W need not be transversal. The subspace W is called

(1) **isotropic** if $W \leq W^\omega$,

(2) **coisotropic** if $W^\omega \leq W$,

(3) **Lagrangian** if $W = W^\omega$,

(4) **symplectic** if $W \cap W^\omega = 0$.

Note that W is isotropic iff ω vanishes on W, and symplectic iff $(W, \omega|_W)$ is a symplectic space.

Lemma 1.1.7

For a subspace $W \leq V$,

$$\dim W + \dim W^\omega = \dim V, \qquad W^{\omega\omega} = W.$$

Proof

Consider $\widehat{\omega}\colon V \to V^*$ which is an isomorphism by non-degeneracy. Let us define

$$\operatorname{ann} W = \{\varphi \in V^* \mid \varphi(W) = 0\}.$$

Because $\widehat{\omega}$ is an isomorphism, we have

$$\operatorname{ann} W = \{\widehat{\omega}(v) \mid \widehat{\omega}(v)(W) = 0\} = \widehat{\omega}(W^\omega)$$

as desired.

Notice that a functional in $\operatorname{ann} W$ is the same as a functional over V/W. Indeed, $\varphi \in \operatorname{ann} W$ iff $W \leq \ker \varphi$ iff we can quotient φ by W. Thus $\operatorname{ann} W \cong (V/W)^*$ which has dimension $\dim V - \dim W$ as desired. ■

Let V be a $2n$-dimensional vector space with basis $u_1, \ldots, u_n, v_1, \ldots, v_n$. Then we can define a symplectic form on V by

$$\omega = \sum_{i=1}^{n} du_i \wedge dv_i,$$

equivalently

$$\omega(u_i, u_j) = \omega(v_i, v_j) = 0, \quad \omega(u_i, v_j) = \delta_{ij}, \qquad [\omega] = J_0.$$

Conversely any basis of a symplectic space which represents the symplectic form like this is called a **symplectic basis**.

Notice that the existence of a symplectic basis is equivalent to the existence of a symplectomorphism $(\mathbb{R}^{2n}, \omega_0) \to (V, \omega)$.

Theorem 1.1.8

Let (V, ω) be a symplectic space. Then V has even dimension, $\dim V = 2n$, and there exists a symplectic basis.

Proof

We induct on $m = \dim V$. For $m = 0$ this is trivial, so suppose $m > 0$. Because ω is non-degenerate there are $u_1, v_1 \in V$ such that $\omega(u_1, v_1) \neq 0$. We can normalize this and assume $\omega(u_1, v_1) = 1$.

Because ω is alternating, this means that u_1, v_1 are linearly independent. Furthermore the subspace W spanned by u_1, v_1 is symplectic as ω is nondegenerate there. Thus $V = W \oplus W^\omega$, and by induction W^ω has even dimension and a symplectic basis. By adding u_1, v_1 to W^ω's symplectic basis we get a symplectic basis of V, and V has even dimension. ■

An immediate corollary of this theorem is the following.

Corollary 1.1.9

Two symplectic spaces V, W are symplectomorphic iff $\dim V = \dim W$.

Corollary 1.1.10

Let V be a $2n$-dimensional vector space and ω a 2-form on V. Then ω is nondegenerate iff its n-fold exterior power ω^n is nonzero.

Proof

If ω is degenerate then there is a $v \neq 0$ such that $v \lrcorner \omega = 0$. Because interior multiplication is an antiderivation:

$$v \lrcorner \omega^n = \omega \wedge (v \lrcorner \omega^{n-1}) + (v \lrcorner \omega) \wedge \omega^{n-1} = \omega \wedge (v \lrcorner \omega^{n-1}).$$

Thus inductively we have $v \lrcorner \omega^n = 0$. Because ω^n is a top-form this means $\omega^n = 0$ (we can see this by extending v to a basis, then ω on that basis is zero).

Conversely if ω is nondegenerate, then by the previous theorem there is a symplectic basis

$$\omega = \sum_i dx_i \wedge dy_i.$$

Then

$$\omega^n = \sum_I dx_{i_1} \wedge dy_{i_1} \wedge \cdots \wedge dx_{i_n} \wedge dy_{i_n},$$

and we may remove Is with repeated indices, so the sum is over permutations of $1, \ldots, n$. Each permutation gives the same product as 2-forms commute, thus

$$\omega^n = n!\, dx_1 \wedge dy_1 \wedge \cdots dx_n \wedge dy_n \neq 0.$$ ■

Proposition 1.1.11

The Lagrangian subspaces of V are precisely the maximal isotropic subspaces. Moreover, every basis of a Lagrangian subspace can be extended to a symplectic basis of V.

Proof

If W is isotropic then $W \leq W^\omega$ meaning $\dim W \leq \frac{1}{2} \dim V$. Since Lagrangian subspaces have dimension equal to $\frac{1}{2} \dim V$, Lagrangian subspaces are maximal. Conversely if W if isotropic and $W \neq W^\omega$, then there is a $v \in W^\omega - W$. Adding v to W results in an isotropic subspace as ω vanishes on $W + v$. Thus maximal isotropic subspaces must have $W = W^\omega$ so that W is Lagrangian.

For the second point it is sufficient to consider $V = \mathbb{R}^{2n}$ with the standard symplectic structure. Let $\Lambda \leq V$ be Lagrangian, then $\Lambda' = J_0 \Lambda$ is symplectic as well. Notice that $\Lambda' \cap \Lambda = 0$ because for $u \in \Lambda$ we have $\omega_0(u, J_0 u) = -u^\top u \neq 0$. Thus $\mathbb{R}^{2n} = \Lambda' \oplus \Lambda$.

Let $u_1, \ldots, u_n$ be a basis for Λ. Consider the map $\widehat{\omega} \colon \Lambda' \to \Lambda^*$. This is injective as its kernel is $\Lambda \cap \Lambda^\omega = 0$, and thus defines an isomorphism. Let $\varphi_1, \ldots, \varphi_n$ be the dual basis of $u_1, \ldots, u_n$, and thus there are $v_1, \ldots, v_n \in \Lambda'$ such that $\varphi_i = \widehat{\omega}(v_i)$.

Because Λ, Λ' are Lagrangian we have that

$$\omega(u_i, u_j) = \omega(v_i, v_j) = 0, \qquad \omega(v_i, u_j) = \varphi_i(u_j) = \delta_{ij}$$

as desired. ■

Proposition 1.1.12

Let $W \leq V$ be coisotropic.

(1) The quotient $\overline{W} = W/W^\omega$ has a unique symplectic structure $\overline{\omega}$ whose pullback under projection $W \to \overline{W}$ agrees with $\omega|_W$.

(2) If $\Lambda \leq V$ is Lagrangian, then

$$\overline{\Lambda} = \frac{(\Lambda \cap W) + W^\omega}{W^\omega} \leq \overline{W}$$

is Lagrangian.

Proof

Let $[w] \in \overline{W}$ denote the coset of w. Notice that $\overline{\omega}$ must satisfy

$$\omega(w, w') = \overline{\omega}([w], [w']),$$

which uniquely determines $\overline{\omega}$, showing uniqueness. Because W is coisotropic, W^ω is an isotropic subspace of W, and so this is well-defined: for $v, v' \in W^\omega$ and $w, w' \in W$,

$$\omega(w + v, w' + v') = \omega(w, w') + \omega(v, w') + \omega(w, v') + \omega(v, v') = \omega(w, w').$$

The middle two terms are zero because $v, v' \in W^\omega$, and $v \in W^\omega \leq W$ so the last term is zero. Furthermore $\overline{\omega}$ is nondegenerate as if $\overline{\omega}([w], \bullet) = 0$, then $w \in W^\omega$ so $[w] = 0$. So we have shown (1).

For (2) we show that

$$\widetilde{\Lambda} = (\Lambda \cap W) + W^\omega \leq V$$

is Lagrangian. Indeed,

$$\widetilde{\Lambda}^\omega = (\Lambda \cap W)^\omega \cap W = (\Lambda^\omega + W^\omega) \cap W = \Lambda \cap W + W^\omega = \widetilde{\Lambda}.$$

And $\widetilde{\Lambda}$ quotients to $\overline{\Lambda}$, and $\widetilde{\Lambda}^\omega$ quotients to $\overline{\Lambda}^\omega$, which are then equal. ■

1.2 The symplectic group

Recall that we have defined a groupoid whose objects are symplectic spaces, and whose isomorphisms are the symplectomorphisms. And corollary 1.1.9 says that the dimension of a symplectic space is its only significant invariant. Let us denote by

$$\mathrm{Sp}(V, \omega) = \{\Psi \in \mathrm{GL}(V) \mid \Psi^* \omega = \omega\}$$

the automorphism group of (V, ω), called the **symplectic group**.

By proposition 1.1.5, $\mathrm{Sp}(V, \omega)$ is a Lie group. Following this proposition, we have the following result.

Theorem 1.2.1

The symplectic group $\mathrm{Sp}(V, \omega)$ is a Lie group with Lie algebra

$$\mathfrak{sp}(V, \omega) = \{T \in \mathrm{end}(V) \mid \omega(A\bullet, \bullet) + \omega(\bullet, A\bullet) = 0\}.$$

We can understand this group by focusing on the case of $(\mathbb{R}^{2n}, \omega_0)$. Then $\mathrm{Sp}(2n) = \mathrm{Sp}(\mathbb{R}^{2n}, \omega_0)$ may be thought of as a Lie group of matrices $\Psi \in \mathrm{Mat}_{2n}$ satisfying $\Psi^\top J_0 \Psi = J_0$. Such matrices are called **symplectic**.

Lemma 1.2.2

Symplectic matrices have determinant one.

Proof

By corollary 1.1.10, ω_0^n is nonzero and thus a volume form on $\mathbb{R}^{2n}$. Furthermore a symplectic matrix Ψ satisfies $\Psi^* \omega_0 = \omega_0$ and thus $\Psi^* \omega_0^n = \omega_0^n$. Because this is a volume form, $\det \Psi = 1$. ■

We identify $\mathbb{R}^{2n}$ with $\mathbb{C}^n$ by identifying $(x, y) \in \mathbb{R}^{2n}$ with $x + iy \in \mathbb{C}^n$. Then J_0 corresponds to multiplication by i in $\mathbb{C}^n$. With this identification, $\mathrm{GL}_n(\mathbb{C})$ is the subgroup of $\mathrm{GL}_{2n}(\mathbb{R})$ consisting of matrices which commute with J_0:

$$\mathrm{GL}_n(\mathbb{C}) = \{A \in \mathrm{GL}_{2n}(\mathbb{R}) \mid J_0 A = A J_0\}.$$

In this identification, a matrix $X + iY \in \mathrm{Mat}_n(\mathbb{C})$ corresponds to the matrix

$$X + iY \in \mathrm{Mat}_n(\mathbb{C}) \leftrightarrow \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix} \in \mathrm{Mat}_{2n})(\mathbb{R}).$$

Notice how the conjugate transpose acts on $\mathrm{Mat}_n(\mathbb{C})$:

$$(X + iY)^\dagger = (X - iY)^\top = \begin{pmatrix} X^\top & Y^\top \\ -Y^\top & X^\top \end{pmatrix} = \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix}^\top.$$

Thus $X + iY \in \mathrm{U}(n)$ iff

$$\begin{pmatrix} X & -Y \\ Y & X \end{pmatrix}^{-1} = \begin{pmatrix} X^\top & Y^\top \\ -Y^\top & X^\top \end{pmatrix},$$

which is iff

$$XX^\top + YY^\top = I, \qquad XY^\top = YX^\top.$$

Notice that

$$i(X + iY)^\dagger (X + iY) = i \iff (X + iY)^\dagger = (X + iY)^{-1} \iff X + iY \in \mathrm{U}(n).$$

Thus we see

$$\mathrm{U}(n) = \mathrm{GL}_n(\mathbb{C}) \cap \mathrm{Sp}(2n).$$

In fact we can extend this result.

Lemma 1.2.3

$$\mathrm{Sp}(2n) \cap \mathrm{O}(2n) = \mathrm{Sp}(2n) \cap \mathrm{GL}_n(\mathbb{C}) = \mathrm{O}(2n) \cap \mathrm{GL}_n(\mathbb{C}) = \mathrm{U}(n).$$

Proof

A matrix $\Psi \in \mathrm{GL}_{2n}(\mathbb{R})$ satisfies

$$\Psi \in \mathrm{GL}_n(\mathbb{C}) \iff \Psi J_0 = J_0 \Psi, \qquad \Psi \in \mathrm{Sp}(2n) \iff \Psi^\top J_0 \Psi = J_0, \qquad \Psi \in \mathrm{O}(2n) \iff \Psi^\top \Psi = I.$$

Note that any of these two conditions imply the third, so we have the first two equalities. By the above discussion, we also get the final. ∎

For a linear operator Ψ we denote its spectrum by $\sigma(\Psi)$; this is its set of eigenvalues.

Lemma 1.2.4

Let $\Psi \in \mathrm{Sp}(2n)$, then $\lambda \in \sigma(\Psi)$ iff $\lambda^{-1} \in \sigma(\Psi)$, and the multiplicities of λ and λ^{-1} are equal. If ± 1 is an eigenvalue of Ψ then it has even multiplicity.

Finally, if $\lambda, \lambda' \in \sigma(\Psi)$ are two (not necessarily distinct) eigenvalues such that $\lambda\lambda' \neq 1$, then $\omega_0(z, z') = 0$ for any eigenvectors of eigenvalues λ, λ' respectively.

Proof

The first statement is due to the fact that Ψ^{-1} is similar to $\Psi^\top$:

$$\Psi^\top = J_0 \Psi^{-1} J_0^{-1},$$

Ψ and $\Psi^\top$ share the same eigenvalues (with the same multiplicities), and λ is an eigenvalue of Ψ iff λ^{-1} is of Ψ^{-1} (and the multiplicities agree).

Thus the total multiplicity of the eigenvalues not equal to ± 1 is even (as they come in pairs). The determinant of Ψ is the product of eigenvalues, as they come in pairs with their inverses this is equal to $(-1)^m$ where m is -1's multiplicity. By lemma 1.2.2 this is equal to 1 so m is even. Because Ψ has dimension $2n$, the multiplicity of 1 must be even as well.

The last statement is because

$$\omega_0(z, z') = \omega_0(\Psi z, \Psi z') = \lambda\lambda'\omega_0(z, z')$$

and since $\lambda\lambda' \neq 1$, $\omega_0(z, z') = 0$. ■

Suppose D is a diagonal matrix, $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, then define

$$D^\alpha = \operatorname{diag}(\lambda_1^\alpha, \ldots, \lambda_n^\alpha)$$

for any real $\alpha \in \mathbb{R}$ where this is defined. Suppose $A \in \operatorname{Mat}_n(\mathbb{R})$ is a diagonalizable matrix with $A = P^{-1}DP$, D diagonal. Then define

$$A^\alpha = P^{-1}D^\alpha P$$

for real α where this is defined. Note that this agrees with A^n for n natural. It is easy to see that this is well-defined (i.e. not dependent on P, D).

If A^α exists and $Av = \lambda v$, then $A^\alpha v = \lambda^\alpha v$. This is clear for diagonal A, and extends to diagonalizable A by extending v to an eigenbasis and using that P.

Note that if A's eigenvalues are all positive, then this is defined for all α. In particular if S is symmetric and positive definite, then its powers exist for all real α.

Lemma 1.2.5

Exponentiation is continuous as a map

$$\operatorname{Sym}_+ \times \mathbb{R} \to \operatorname{Sym}_+, \qquad (S, \alpha) \mapsto S^\alpha$$

where $\operatorname{Sym}_+ \subseteq \operatorname{Mat}_n$ is the subspace of positive-definite symmetric matrices.

Proof

Let $S \in \operatorname{Sym}_+$, and let its minimum and maximum eigenvalues by $\lambda_{\min}, \lambda_{\max}$ respectively. Then we know for all vectors v,

$$\lambda_{\min} |v|^2 \leq v^\top S v \leq \lambda_{\max} |v|^2$$

and these bounds are tight. We claim that there is an open neighborhood $S \in U \subseteq \operatorname{Sym}_+$ and numbers $0 < m < M$ such that every $A \in U$ has eigenvalues bound by m, M.

Let $m = \lambda_{\min}/2$, then for every symmetric matrix A and v of unit norm:

$$v^\top A v = v^\top S v + v^\top (A - S) v \geq \lambda_{\min} - \left| v^\top (A - S) v \right| \geq \lambda_{\min} - |(A - S)v| \geq \lambda_{\min} - |A - S|$$

where $|A - S|$ denotes the operator norm. Thus if $|A - S| < m$ then the right-hand side is equal to $\lambda_{\min}/2$ and thus $v^\top A v \geq \lambda_{\min}/2$; so all of A's eigenvalues are bound from below by m. So all matrices in the ball of radius m around S have eigenvalues bound from below by m.

We can bound this from above too: for unit norm v

$$v^\top A v \leq |A| \leq |S| + |A - S|,$$

and so if we require $|A - S| < 1$ then

$$v^\top A v \leq |S| + 1 = \lambda_{\max} + 1,$$

so matrices in the ball of radius 1 around S have eigenvalues bound from above by $M = \lambda_{\max} + 1$. So the desired open set is

$$U = \{A \in \mathrm{Sym}_+ \mid |A - S| < 1, m\}.$$

Instead of considering exponentiating over arbitrary α, we show it is continuous for $a \leq \alpha \leq b$ for any arbitrary a, b; this is clearly sufficient. Consider the map $f\colon [m, M] \times [a, b] \to \mathbb{R}$ by $f(x, t) = t^x$, this is continuous as $m > 0$. By Stone-Weirestrass, because the codomain is compact, we can approximate f uniformly by a sequence of polynomials $p_k \in \mathbb{R}[x, t]$.

Now suppose $A \in U$, so $A = P^{-1} \operatorname{diag}(\lambda_1, \ldots, \lambda_n)P$, and because p_k are polynomial

$$p_k(x, A) = P^{-1} \operatorname{diag}(p_k(x, \lambda_1), \ldots, p_k(x, \lambda_n))P.$$

Thus

$$A^\alpha - p_k(\alpha, A) = P^{-1} \operatorname{diag}(\ldots, \lambda_i - p_k(\alpha, \lambda_i), \ldots)P.$$

Because $A \in U$ the λ_i fall in $[m, M]$, and so under the operator norm

$$|A^\alpha - p_k(\alpha, A)| = \max_i |\lambda_i - p_k(\alpha, \lambda_i)| \leq \max_{t \in [m,M]} |t^\alpha - p_k(\alpha, t)|.$$

And so we see that the map $F\colon U \times [a, b] \to \mathrm{Sym}_+$, $F(A, \alpha) = A^\alpha$, satisfies for all A, α

$$|F(A, \alpha) - p_k(\alpha, A)| \leq \max_{t \in [m,M], \alpha \in [a,b]} |t^\alpha - p_k(\alpha, t)|,$$

so that $(\alpha, A) \mapsto p_k(\alpha, A)$ uniformly converges to F. Because p_k are all polynomial and thus continuous, so too is F. ■

Lemma 1.2.6

If $S \in \mathrm{Sp}(2n)$ is symmetric and positive definite, then $P^\alpha \in \mathrm{Sp}(2n)$ for all $\alpha \in \mathbb{R}$.

Proof

As S is symmetric and positive definite we can decompose $\mathbb{R}^{2n}$ into eigenspaces

$$\mathbb{R}^{2n} = \bigoplus_{i=1}^{k} E_i, \qquad E_i = \ker(\lambda_i I - S),$$

for $\lambda_i > 0$ the eigenvalues. For $z, w \in \mathbb{R}^{2n}$ let their projections into E_i be denoted z_i, w_i respectively. Then because S is symplectic

$$\omega_0(z_i, w_j) = \omega_0(Pz_i, Pw_j) = \lambda_i \lambda_j \omega_0(z_i, w_j).$$

So either $\lambda_i \lambda_j = 1$ or $\omega_0(z_i, w_j) = 0$, and thus Then

$$\omega_0(P^\alpha z, P^\alpha w) = \sum_{i,j}^{k} (\lambda_i \lambda_j)^\alpha \omega_0(z_i, w_j) = \sum_{i,j}^{k} \omega_0(z_i, w_j) = \omega_0(z, w)$$

so that $P^\alpha \in \mathrm{Sp}(2n)$. ■

Proposition 1.2.7

(1) $\mathrm{U}(n)$ is a maximal compact subgroup of $\mathrm{Sp}(2n)$.

(2) The inclusion $\mathrm{U}(n) \to \mathrm{Sp}(2n)$ is a strong deformation retract. In particular $\mathrm{Sp}(2n)$ is connected.

Proof

We first prove (2). Define

$$H\colon [0,1] \times \mathrm{Sp}(2n) \to \mathrm{Sp}(2n), \qquad H(t,\Psi) = \Psi(\Psi^\top\Psi)^{-t/2}.$$

This is continuous by lemma 1.2.5, as $\Psi^\top\Psi$ is positive definite for any non-singular Ψ. This is well-defined by the previous lemma, $H(\Psi,t) \in \mathrm{Sp}(2n)$ for all t.

Now

$$H_0 = \mathrm{id}, \qquad H_t|_{\mathrm{U}(n)} = \mathrm{id}$$

where the latter is because $\Psi^\top\Psi = I$ for unitary Ψ. And for $\Psi \in \mathrm{Sp}(2n)$, $H_1(\Psi)$ is orthogonal:

$$H_1(\Psi)^\top H_1(\Psi) = (\Psi^\top\Psi)^{-1/2}\Psi^\top\Psi(\Psi^\top\Psi)^{-1/2} = I.$$

Because $\mathrm{Sp}(2n) \cap \mathrm{O}(2n) = \mathrm{U}(n)$, we have $H_1(\mathrm{Sp}(2n)) = \mathrm{U}(n)$. Thus H is indeed a strong deformation retract. And so $\mathrm{Sp}(2n)$ and $\mathrm{U}(n)$ are homotopy equivalent, and $\mathrm{U}(n)$ is connected so therefore $\mathrm{Sp}(2n)$ is as well.

We show (1) now. Suppose G is a compact subgroup of $\mathrm{Sp}(2n)$ containing $\mathrm{U}(n)$, but not equal to $\mathrm{U}(n)$. Then let $\Psi \in G - \mathrm{U}(n)$, and so $H_1(\Psi) = \Psi(\Psi^\top\Psi)^{-1/2} \in \mathrm{U}(n) \subseteq G$. Thus the positive definite symmetric matrix $P = \Psi^\top\Psi$ belongs to $G - \mathrm{U}(n)$. Therefore it must have an eigenvalue greater than 1: its eigenvalues come in pairs λ, λ^{-1}, so otherwise $P = I$. But then the sequence $P, P^2, P^3, \ldots$ in G has no convergent subsequence, contradicting G's compactness. ■

Remark 1.2.8

Every symplectic matrix $\Psi \in \mathrm{Sp}(2n)$ can be uniquely written as $\Psi = UP$ where $U \in \mathrm{U}(n)$ and P is a symmetric, positive definite, symplectic matrix. Existence is by

$$U = \Psi(\Psi^\top\Psi)^{-1/2}, \qquad P = (\Psi^\top\Psi)^{1/2}.$$

Uniqueness is because the only symmetric positive definite unitary matrix is I (because its eigenvalues come in pairs, and thus must all be 1). If $U \in \mathrm{U}(n)$ can be written as $U = U'P'$, then $P = U(U')^{-1}$ is unitary as well, and thus $P = I$ so $U = U'$. So this is true for unitary matrices, and thus if $UP = U'P'$ we have $U = U'P'P^{-1}$ so $U = U'$ and $P'P^{-1} = I$ meaning $P = P'$. This shows uniqueness.

This decomposition is called **symplectic polar decomposition**.

Proposition 1.2.9

The complex determinant map $\det_{\mathbb{C}}\colon \mathrm{U}(n) \to S^1$ induces an isomorphism of fundamental groups $\pi_1(\mathrm{U}\,n) \cong \pi_1(S^1) \cong \mathbb{Z}$.

Proof

The determinant map is a fibration with typical fiber $\mathrm{SU}(n)$. Indeed for every $e^{i\theta} \in S^1$ we can consider the open subset W of S^1 consisting of numbers whose arguments are between $\theta - \varepsilon$ and $\theta + \varepsilon$. Then $\det_{\mathbb{C}}^{-1}(W) \to \mathrm{SU}(n) \times W$ given by $\Psi \mapsto (e^{i\varphi/n}\Psi, e^{i\varphi})$ where $e^{i\varphi} = \det\Psi$ is a homeomorphism with inverse $(\Phi, e^{i\varphi}) \to e^{i\varphi/n}\Phi$. It also commutes with $\det_{\mathbb{C}}$ because $\det(e^{i\varphi/n}\Phi) = e^{i\varphi}$.

Thus there is an exact sequence

$$\pi_1(\mathrm{SU}(n)) \to \pi_1(\mathrm{U}\, n) \to \pi_1(S^1) \to \pi_0(\mathrm{SU}(n)).$$

It is sufficient then to show that $\mathrm{SU}(n)$ is simply connected, and we get the desired result.

It is trivial that $\mathrm{SU}(n)$ is simply connected for $n = 1$, as $\mathrm{SU}(1) = \{1\}$. For $n \geq 2$ the map $\mathrm{SU}(n) \to S^{2n-1}$ which maps $U \in \mathrm{SU}(n)$ to its first column is a fibration with typical fiber $\mathrm{SU}(n-1)$. And so we have an exact sequence

$$\pi_1(\mathrm{SU}(n-1)) \to \pi_1(\mathrm{SU}\, n) \to \pi_1(S^{2n-1}) \to \pi_0(\mathrm{SU}(n-1)).$$

By induction this means $\pi_1(\mathrm{SU}\, n) \cong \pi_1(S^{2n-1}) = 1$ because $2n - 1 > 1$. ■

Because $\mathrm{U}(n)$ is a deformation retract of $\mathrm{Sp}(2n)$, they have the same fundamental group so $\pi_1(\mathrm{Sp}(2n)) \cong \mathbb{Z}$. We now construct an explicit isomorphism.

Recall that the **degree** of a loop $\gamma\colon \mathbb{R}/\mathbb{Z} \to S^1$ is defined to be its value in $\mathbb{Z}$ under the usual isomorphism $\pi_1(S^1) \cong \mathbb{Z}$. Equivalently, we lift γ to $\widetilde{\gamma}\colon [0,1] \to \mathbb{R}$ (because covering spaces have the path lifting property), and define

$$\deg \gamma = \widetilde{\gamma}(1) - \widetilde{\gamma}(0).$$

Theorem 1.2.10

There exists a unique function μ, called the **Maslov index**, which assigns an integer $\mu(\Psi) \in \mathbb{Z}$ to every loop $\Psi\colon \mathbb{R}/\mathbb{Z} \to \mathrm{Sp}(2n)$ and satisfies the following axioms:

(1) Homotopy: two loops are homotopic iff they have the same Maslov index.

(2) Product: the Maslov index of the product of two loops is the sum of their Maslov indices:

$$\mu(\Psi_1\Psi_2) = \mu(\Psi_1) + \mu(\Psi_2).$$

Note here we are taking the pointwise product of loops of matrices: $(\Psi_1\Psi_2)(t) = \Psi_1(t)\Psi_2(t)$.

(3) Direct sum: if $n = n' + n''$, we may identify $\mathrm{Sp}(2n') \oplus \mathrm{Sp}(2n'')$ as a subgroup of $\mathrm{Sp}(2n)$. Then for loops Ψ', Ψ'' in $\mathrm{Sp}(2n'), \mathrm{Sp}(2n'')$ respectively,

$$\mu(\Psi' \oplus \Psi'') = \mu(\Psi') + \mu(\Psi'').$$

(4) Normalization: the loop $\Psi\colon \mathbb{R}/\mathbb{Z} \to \mathrm{Sp}(2)$ defined by

$$\Psi(t) = \begin{pmatrix} \cos(2\pi t) & -\sin(2\pi t) \\ \sin(2\pi t) & \cos(2\pi t) \end{pmatrix}$$

has Maslov index 1.

Note:

We identify $\mathrm{Sp}(2n') \oplus \mathrm{Sp}(2n'')$ with a subgroup of $\mathrm{Sp}(2n)$ by

$$\begin{pmatrix} X & Y \\ Z & W \end{pmatrix} \oplus \begin{pmatrix} X' & Y' \\ Z' & W' \end{pmatrix} = \left(\begin{array}{cc|cc} X & & Y & \\ & X' & & Y' \\ \hline Z & & W & \\ & Z & & W' \end{array}\right).$$

Proof

Define $\rho\colon \mathrm{Sp}(2n) \to S^1$ by mapping $\Psi \in \mathrm{Sp}(2n)$ (not a loop) to

$$\rho(\Psi) = \det_{\mathbb{C}}(X + iY), \qquad \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix} = \Psi(\Psi^\top \Psi)^{-1/2} \in \mathrm{U}(n).$$

Then define the Maslov index to be

$$\mu(\Psi) = \deg(\rho \circ \Psi).$$

Equivalently $\mu(\Psi) = \alpha(1) - \alpha(0)$ where $\alpha\colon I \to \mathbb{R}$ is a continuous lift of $\rho \circ \Psi$, meaning

$$\exp(2\pi i \alpha(t)) = \det_{\mathbb{C}}(X(t) + iY(t)).$$

Clearly this definition depends only on the homotopy class of Ψ, and by above we know $\rho\colon \mathrm{Sp}(2n) \to S^1$ defines an isomorphism of the fundamental groups. This isomorphism is μ, so we have axiom (1). The product axiom is true for loops of unitary matrices, because $\rho(\Psi) = \det_{\mathbb{C}} \Psi$, so if α_i is a lift of $\rho\Psi_i$ then

$$\det_{\mathbb{C}}(\Psi_1\Psi_2) = \det_{\mathbb{C}} \Psi_1 \cdot \det_{\mathbb{C}} \Psi_2 = \exp(2\pi i \alpha_1(t)) \exp(2\pi i \alpha_2(t)) = \exp(2\pi i(\alpha_1 + \alpha_2)(t)).$$

Thus $\alpha_1 + \alpha_2$ is a lift of $\rho(\Psi_1\Psi_2)$, so $\mu(\Psi_1\Psi_2) = \mu(\Psi_1) + \mu(\Psi_2)$. Because a loop of symplectic matrices is homotopic to a loop of unitary matrices, we have the desired result.

The last two axioms are easily verified. Thus the existence is seen, and uniqueness can be shown as well. ■

1.3 Lagrangian subspaces

For a symplectic space (V, ω), let $\mathcal{L}(V, \omega)$ denote the set of its Lagrangian subspaces. We denote

$$\mathcal{L}(n) = \mathcal{L}(\mathbb{R}^{2n}, \omega_0).$$

We can characterize the Lagrangian subspaces of $\mathbb{R}^{2n}$ more explicitly. Firstly a Lagrangian subspace of $\mathbb{R}^{2n}$ has dimension n, and is thus the image of some full-rank linear transform $\mathbb{R}^n \to \mathbb{R}^{2n}$, or equivalently a $2n \times n$-matrix of rank n. We can then determine when such an image is Lagrangian.

Proposition 1.3.1

Let $X, Y \in \mathrm{Mat}_n$ be matrices, and define $\Lambda \leq \mathbb{R}^{2n}$ by

$$\Lambda = \operatorname{im} Z, \qquad Z = \begin{pmatrix} X \\ Y \end{pmatrix}.$$

Then $\Lambda \in \mathcal{L}(n)$ iff Z has rank n and $X^\top Y = Y^\top X$.

In particular the graph $\Lambda = \{(x, Ax) \mid x \in \mathbb{R}^n\}$ of some matrix $A \in \mathrm{Mat}_n$ is Lagrangian iff A is symmetric.

Proof

It is sufficient to consider when ω_0 is trivial on Λ. Let $z = (Xu, Yu)$ and $z' = (Xu', Yu')$ in Λ, then

$$\omega_0(z, z') = \begin{pmatrix} (Xu)^\top & (Yu)^\top \end{pmatrix} \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix} \begin{pmatrix} Xu' \\ Yu' \end{pmatrix} = u^\top (X^\top Y - Y^\top X) u'.$$

This is zero for all u, u' iff $X^\top Y = Y^\top X$, as desired.

The second claim is because $\Lambda = \operatorname{im}(I, A)$, which has rank n. Thus it is Lagrangian iff $A^\top = A$. ■

A matrix $Z \in \mathbb{R}^{2n \times n}$ of the form

$$Z = \begin{pmatrix} X \\ Y \end{pmatrix}, \qquad X^\top Y = Y^\top X$$

is called a **Lagrangian frame**. Notice that Z's columns form an orthonormal basis for $\Lambda = \operatorname{im} Z$ iff $U = X + iY$ is unitary. Indeed Z's columns are orthonormal iff

$$I = Z^\top Z = X^\top X + Y^\top Y.$$

And U is unitary iff

$$I = U^\dagger U = (X^\top - iY^\top)(X + iY) = X^\top X + Y^\top Y + i(X^\top Y - Y^\top X) = X^\top X + Y^\top Y.$$

A Lagrangian frame Z for which $U = X + iY$ is unitary is called a **unitary Lagrangian frame**. Every Lagrangian subspace has a unitary Lagrangian frame as we can take an orthonormal basis and put it in a matrix.

Let us define the Lagrangian subspace

$$\Lambda_{\mathsf{hor}} = \left\{(x, 0) \in \mathbb{R}^{2n} \;\middle|\; x \in \mathbb{R}^n\right\} = \operatorname{im} \begin{pmatrix} I \\ 0 \end{pmatrix}.$$

Lemma 1.3.2

(1) If $\Lambda \in \mathcal{L}(n)$ and $\Psi \in \mathrm{Sp}(2n)$, then $\Psi\Lambda \in \mathcal{L}(n)$.

(2) For any two Lagrangian subspaces $\Lambda, \Lambda' \in \mathcal{L}(n)$, there is a symplectic matrix $\Psi \in \mathrm{U}(n)$ such that $\Lambda' = \Psi\Lambda$.

(3) $\mathcal{L}(n)$ can be seen to be naturally isomorphic to $\mathrm{U}(n)/\mathrm{O}(n)$, and thus carries the natural structure of a manifold of dimension $n(n+1)/2$.

Proof

(1) is clear. For (2) let $\Lambda \leq \mathbb{R}^{2n}$ be given by a unitary Lagrangian frame $Z = (X, Y)$ and define

$$\Psi = X + iY = \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix} \in \mathrm{Sp}(2n) \cap \mathrm{O}(n).$$

Then

$$\Psi\Lambda_{\mathsf{hor}} = \Lambda,$$

and thus we have shown (2).

A unitary Lagrangian frame of Λ is uniquely determined up to choice of orthonormal basis. Altering the basis amounts to multiplication on the right by an orthonormal matrix, so we see that there is a bijection $\mathrm{U}(n)/\mathrm{O}(n) \cong \mathcal{L}(n)$. The dimension of $\mathrm{U}(n)$ is n^2, and the dimension of $\mathrm{O}(n)$ is $n(n-1)/2$, so $\dim \mathcal{L}(n) = n(n+1)/2$. ■

We can show that $\mathcal{L}(n) \cong \mathrm{U}(n)/\mathrm{O}(n)$ has a fundamental group isomorphic to the integers. This isomorphism is also called the **Maslov index**.

Theorem 1.3.3

There exists a unique function, the **Maslov index**, μ which assigns to each loop $\Lambda\colon \mathbb{R}/\mathbb{Z} \to \mathcal{L}(n)$ an integer and satisfies the following axions:

(1) homotopy: two loops in $\mathcal{L}(n)$ are homotopic iff they have the same Maslov index.

(2) product: for any two loops $\Lambda\colon \mathbb{R}/\mathbb{Z} \to \mathcal{L}(n)$ and $\Psi\colon \mathbb{R}/\mathbb{Z} \to \mathrm{Sp}(2n)$, we have

$$\mu(\Psi\Lambda) = \mu(\Lambda) + 2\mu(\Psi).$$

(3) direct sum: if $n = n' + n''$, then we can identify $\mathcal{L}(n') \oplus \mathcal{L}(n'')$ as a submanifold of $\mathcal{L}(n)$ and then

$$\mu(\Lambda' \oplus \Lambda'') = \mu(\Lambda') + \mu(\Lambda'').$$

(4) zero: a constant loop $\Lambda(t) = \Lambda_0$ has Maslov index zero.

(5) normalization: the loop $\Lambda\colon \mathbb{R}/\mathbb{Z} \to \mathcal{L}(1)$ given by

$$\Lambda(t) = e^{\pi i t}\mathbb{R} \leq \mathbb{C} = \mathbb{R}^2$$

has Maslov index one.

Proof

An orthonormal matrix has determinant ± 1, thus the map

$$\rho\colon \mathcal{L}(n) \cong \mathrm{U}(n)/\mathrm{O}(n) \to S^1, \qquad \rho([U]) = \det_{\mathbb{C}} U^2$$

is well-defined. Here $[U] \in \mathrm{U}(n)/\mathrm{O}(n)$ denotes the coset of the unitary matrix $U = X + iY$, which in turn represents the Lagrangian subspace $\Lambda = \mathrm{im}(X, Y)$.

Define the Maslov index to be the induced map $\pi_1(\mathcal{L}(n)) \to \pi_1(S^1) \xrightarrow{\sim} \mathbb{Z}$. That is for a loop Λ,

$$\mu(\Lambda) = \deg(\rho \circ \Lambda) = \alpha(1) - \alpha(0)$$

where $\alpha\colon I \to \mathbb{R}$ is a lift of $\rho \circ \Lambda$, meaning

$$e^{\pi i \alpha(t)} = \det_{\mathbb{C}}(U(t)), \qquad \Lambda = [U].$$

The Maslov index then depends only on the homotopy class of Λ. We show it satisfies the product axiom: first note that if Ψ is unitary $\rho(\Psi\Lambda) = \rho(\Psi)^2\rho(\Lambda)$ ($\rho(\Psi)$ corresponds to the ρ in theorem 1.2.10). And so the degree of this is

$$\deg(\rho(\Psi\Lambda)) = \deg \rho(\Lambda) + 2 \deg \rho(\Psi) = \mu(\Lambda) + 2\mu(\Psi).$$

If Ψ is not unitary, it is homotopic to a loop of unitary matrices, and the result follows.

We now show the converse direction in the homotopy axiom. Suppose that Λ_0, Λ_1 are two loops of Lagrangian subspaces with the same Maslov index, we must show Λ_0, Λ_1 are homotopic. By the previous lemma and the product axiom, we may assume $\Lambda_j(0) = \Lambda_j(1) = \Lambda_{\mathsf{hor}}$. Lift $\Lambda_j(t)$ to $U_j(t) = X_j(t) + iY_j(t) \in \mathrm{U}(n)$ such that $U_0(0) = U_1(0) = I$ (as these represent Λ_{hor}). We may multiply U_j by right multiplication with orthogonal matrices to get $U_0(1) = U_1(1) = I$ as well. Then $U(t) = U_1(t)U_0(t)^{-1}$ forms a loop of unitary matrices.

Note that $\mu(\Lambda_0) = \mu(\Lambda_1)$ means $2\mu(U_0) = 2\mu(U_1)$, and thus U_0, U_1 are homotopic. Thus Λ_0, Λ_1 are homotopic.

The normalization axiom follows from the other axioms. Note that the loop $\Lambda(t) = e^{\pi i t}\mathbb{R}$ satisfies

$$\Lambda(t) \oplus \Lambda(t) = \Psi(t)\Lambda_0$$

where $\Lambda_0 = \mathbb{R}^2 \leq \mathbb{C}^2$ is constant and Ψ is the loop

$$\Psi(t) = e^{2\pi i t}X(t) = e^{\pi i t}\begin{pmatrix} \cos(\pi t) & -\sin(\pi t) \\ \sin(\pi t) & \cos(\pi t) \end{pmatrix}.$$

Here $X(t)$ has constant determinant 1 and so $\det \Psi = e^{2\pi i t}$ lifts to $\alpha(t) = t$ and thus $\mu(\Psi) = 1$. Thus we have

$$2\mu(\Lambda) = \mu(\Lambda \oplus \Lambda) = \mu(\Lambda_0) + 2\mu(\Psi) = 2 \implies \mu(\Lambda) = 1. \qquad \blacksquare$$

1.4 The affine nonsqueezing theorem

Define an **affine symplectomorphism** of $\mathbb{R}^{2n}$ to be a map $\psi\colon \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ of the form

$$\psi(z) = \Psi(z) + z_0, \qquad \Psi \in \mathrm{Sp}(2n),\ z_0 \in \mathbb{R}^{2n}.$$

Let $\mathrm{ASp}(2n)$ denote the set of affine symplectomorphisms, this is a group as

$$\psi \circ \psi'(z) = \Psi \circ \Psi'(z) + \Psi(z_0') + z_0, \qquad \psi^{-1}(z) = \Psi^{-1}(z) - \Psi^{-1}(z_0)$$

are affine symplectomorphisms.

The affine nonsqueezing theorem says that a ball in $\mathbb{R}^{2n}$ can only be embedded into a symplectic cylinder if the ball's radius is smaller than the cylinder. If we let $e_1, \dots, e_n, f_1, \dots, f_n$ denote the standard basis of $\mathbb{R}^{2n}$, then the symplectic cylinder of radius $R > 0$ is

$$Z^{2n}(R) = B^2(R) \times \mathbb{R}^{2n-2} = \left\{ z \in \mathbb{R}^{2n} \;\middle|\; \langle z, e_1\rangle^2 + \langle z, f_1\rangle^2 \le R^2 \right\} \subseteq \mathbb{R}^{2n}.$$

Theorem 1.4.1 (The affine nonsqueezing theorem)

Let $\psi \in \mathrm{ASp}(2n)$ be an affine symplectomorphism such that $\psi(B^{2n}(r)) \subseteq Z^{2n}(R)$. Then $r \le R$.

Proof

We may assume that $r = 1$ and let $\psi(z) = \Psi(z) + z_0$. Then define

$$u = \Psi^\top e_1, \qquad v = \Psi^\top f_1, \qquad a = \langle e_1, z_0\rangle, \qquad b = \langle f_1, z_0\rangle.$$

Because $\Psi \in \mathrm{Sp}(2n)$ we have

$$\omega_0(u, v) = \omega_0(\Psi^\top e_1, \Psi^\top f_1) = \omega_0(e_1, f_1) = 1.$$

The condition $\psi(B^{2n}(1)) \subseteq Z^{2n}(R)$ is equivalent to

$$\sup_{|z|=1} \left(\langle e_1, \psi(z)\rangle^2 + \langle f_1, \psi(z)\rangle^2 \right) = \sup_{|z|=1} \left((\langle u, z\rangle + a)^2 + (\langle v, z\rangle + b)^2 \right) \le R^2.$$

Because $1 = \omega_0(u, v) \le |u|\,|v|$ (as $\omega_0(x, y) = \langle x, J_0 y\rangle$ and J_0 is norm-preserving), one of u, v must have norm at least 1. Without loss of generality $|u| \ge 1$, and then let $z = \pm u/|u|$ whose sign agrees with a's. Then

$$(\langle u, z\rangle + a)^2 \ge |u|^2 \ge 1,$$

which then implies that $1 \le R^2$ and so $1 \le R$ as desired. ∎

Because $B^{2n}(R) \subseteq Z^{2n}(R)$, we have that

$$R = \sup\left\{ r > 0 \;\middle|\; \text{there exists an affine/linear symplectomorphism } \psi \text{ such that } \psi(B^{2n}(r)) \subseteq Z^{2n}(R) \right\}.$$

Either affine or linear in the above supremum gives the same result.

Call a set $B \subseteq \mathbb{R}^{2n}$ an **linear** (resp. **affine**) **symplectic ball** of radius r if it is linearly (resp. affinely) symplectomorphic to $B^{2n}(r)$. Similarly $Z \subseteq \mathbb{R}^{2n}$ is a **linear** (resp. **affine**) **symplectic cylinder** of radius R if it is linearly (resp. affinely symplectomorphic to $Z^{2n}(R)$.

The radius of a symplectic ball or cylinder is invariant. This follows from the above theorem: if B is symplectomorphic to $B^{2n}(r)$ and $B^{2n}(r')$, then $B^{2n}(r), B^{2n}(r')$ are symplectomorphic. By the above theorem this means that $r = r'$. And the radius of a symplectic cylinder is

$$R = \sup\left\{ r > 0 \;\middle|\; \text{there exists an affine/linear symplectomorphism } \psi \text{ such that } \psi(B^{2n}(r)) \subseteq Z^{2n}(R) \right\}.$$

A matrix $\Psi \in \mathrm{Mat}_{2n}$ has the **linear nonsqueezing property** if for every linear symplectic ball B of radius r and linear symplectic cylinder Z of radius R

$$\Psi B \subseteq Z \implies r \le R.$$

By The affine nonsqueezing theorem any symplectic matrix has the linear nonsqueezing property. Call a matrix **anti-symplectic** if $\Psi^* \omega_0 = -\omega_0$. The affine nonsqueezing theorem clearly holds for anti-symplectic matrices as well.

Theorem 1.4.2 (The affine rigidity theorem)

Let $\Psi \in \mathrm{Mat}_{2n}$ be nonsingular such that Ψ, Ψ^{-1} have the linear nonsqueezing property. Then Ψ is either symplectic or anti-symplectic.

Proof

Suppose Ψ is neither symplectic nor anti-symplectic. Then neither is $\Psi^\top$ so there are $u, v \in \mathbb{R}^{2n}$ such that

$$\omega_0(\Psi^\top u, \Psi^\top v) \neq \pm\omega_0(u, v).$$

We may assume that neither side is zero, and potentially by replacing Ψ with Ψ^{-1}, $\left|\omega_0(\Psi^\top u, \Psi^\top v)\right| < |\omega_0(u, v)|$. We can rescale u so that $\omega_0(u, v) = 1$ so that

$$0 < \lambda^2 = \left|\omega_0(\Psi^\top u, \Psi^\top v)\right| < \omega_0(u, v) = 1.$$

Thus $\operatorname{span}\{u, v\}$ is a symplectic subspace of which u, v is a symplectic basis, so we can extend it to a symplectic basis of all of $\mathbb{R}^{2n}$ by considering its symplectic complement. So let $u_1, v_1, \ldots, u_n, v_n$ be a symplectic basis of $\mathbb{R}^{2n}$ with $u = u_1, v = v_1$. Now let $u_1' = \lambda^{-1}\Psi^\top u$ and $v_1' = \pm\lambda^{-1}\Psi^\top v$ where the sign is so that $\omega_0(u_1', v_1') = 1$. Extend this to a symplectic basis $u_1', v_1', \ldots, u_n', v_n'$ of $\mathbb{R}^{2n}$.

Define $\Phi, \Phi' \in \mathrm{Sp}(2n)$ to be the symplectic matrices which map the standard basis $e_1, \ldots, f_n$ to $u_1, \ldots, v_n$ and $u_1', \ldots, v_n'$ respectively. Then $A = (\Phi')^{-1}\Psi^\top\Phi$ satisfies

$$Ae_1 = \lambda e_1, \qquad Af_1 = \pm\lambda f_1.$$

Thus $A^\top$ maps the unit ball $B^{2n}(1)$ in $Z^{2n}(\lambda)$: if $|z| \leq 1$ then

$$\left\langle A^\top z, e_1\right\rangle^2 + \left\langle A^\top z, f_1\right\rangle^2 = \langle z, Ae_1\rangle^2 + \langle z, Af_1\rangle^2 = \lambda^2(\langle z, e_1\rangle^2 + \langle z, f_1\rangle^2) \leq \lambda^2.$$

Because $\lambda < 1$, this means that $A^\top$ does not have the linear squeezing property. Because Φ, Φ' are symplectic this means that Ψ does not have the linear squeezing property. ■

Let $A \subseteq \mathbb{R}^{2n}$ be an arbitrary subset. Define its **linear symplectic width** to be

$$w_L(A) = \sup\left\{\pi r^2 \;\middle|\; \psi(B^{2n}(r)) \subseteq A \text{ for some } \psi \in \mathrm{ASp}(\mathbb{R}^{2n})\right\}.$$

The linear symplectic width has the following properties:

(1) **monotonicity**: if $\psi(A) \subseteq B$ for $\psi \in \mathrm{ASp}(\mathbb{R}^{2n})$ then $w_L(A) \leq w_L(B)$.

(2) **conformality**: $w_L(\lambda A) = \lambda^2 w_L(A)$.

(3) **nontriviality**: $w_L(B^{2n}(r)) = w_L(Z^{2n}(r)) = \pi r^2$.

(1) is obvious, (2) is because $\psi(B^{2n}(r)) \subseteq A$ iff $\psi(B^{2n}(\lambda r)) \subseteq \lambda A$. (3) is due to The affine nonsqueezing theorem: $B^{2n}(r')$ embeds into $B^{2n}(r), Z^{2n}(r)$ iff $r' \leq r$ so $w_L(B^{2n}(r)) = \pi r^2$.

Lemma 1.4.3

Anti-symplectic linear maps have determinant $(-1)^n$. Furthermore anti-symplectic affine maps preserve linear symplectic width.

Proof

If $\Psi^*\omega_0 = -\omega_0$ then $\Psi^*\omega_0^n = (-1)^n\omega_0$. Because ω_0^n is a volume form, this means that $\det\Psi = (-1)^n$ as desired.

For the second point note that the composition of two anti-symplectic maps is symplectic, of a symplectic with an anti-symplectic is anti-symplectic. Define $D = \operatorname{diag}(I, -I)$; this is anti-symplectic and $D^2 = I$. Thus Ψ is anti-symplectic iff ΨD is symplectic. Furthermore D is orthonormal and thus $D(B^{2n}(r)) = B^{2n}(r)$.

This shows that $w_L(A)$ can be defined by taking the supremum over $\psi(B^{2n}(r)) \subseteq A$ where ψ is an anti-symplectic affine map. Thus w_L is monotonic with respect to anti-symplectic affine maps, and we have shown the second claim. ■

An **ellipsoid** is a subset $E \subseteq \mathbb{R}^{2n}$ of the form $\left\{z \in \mathbb{R}^{2n} \;\middle|\; Q(z) \leq 1\right\}$ where Q is any positive definite quadratic form. That is $Q(z) = z^\top \overline{Q} z$ for a symmetric positive-definite $\overline{Q} \in \operatorname{Sym}_+(2n)$. Notice that if $\Psi \in \operatorname{Mat}_{2n}$ is nonsingular then ΨE is an ellipsoid as well:

$$\Psi(E) = \left\{z \in \mathbb{R}^{2n} \;\middle|\; Q(\Psi^{-1} z) \leq 1\right\},$$

and $Q \circ \Psi^{-1}$ is a positive definite quadratic form.

Theorem 1.4.4

Let $\Psi \in \operatorname{Mat}_{2n}$ be a linear map. Then the following are equivalent.

(1) Ψ preserves the linear symplectic width of all ellipsoids.

(2) Ψ is either symplectic or anti-symplectic.

Proof

As shown above, symplectic and anti-symplectic maps preserve all widths, so (2) implies (1). So assume (2) and we will show that Ψ has the linear nonsqueezing property. Let B be a linear symplectic ball of radius r and Z a linear symplectic cylinder of radius R such that $\Psi B \subseteq Z$. Then by hypothesis

$$\pi r^2 = w_L(B) = w_L(\Psi B) \leq w_L(Z) = \pi R^2 \Longrightarrow r \leq R$$

as desired.

A symplectic ball cannot be embedded into a subspace of dimension $< 2n$, so if Ψ were singular then $w_L(\Psi B^{2n}(1)) = 0$, contradicting (1). Thus Ψ is nonsingular, and Ψ^{-1} satisfies (1) because

$$w_L(\Psi^{-1} E) = w_L(\Psi \Psi^{-1} E) = w_L(E)$$

as $\Psi^{-1} E$ is an ellipsoid.

Thus both Ψ and Ψ^{-1} have the linear nonsqueezing property, so by The affine rigidity theorem Ψ is symplectic or anti-symplectic. ■

Lemma 1.4.5

Let $g \colon V \times V \to \mathbb{R}$ be an inner product on a symplectic space. Then there exists a basis $u_1, \ldots, u_n, v_1, \ldots, v_n$ of V which is both g-orthogonal and ω-standard. More so, the basis can be chosen so that $g(u_j, u_j) = g(v_j, v_j)$ for all j.

Proof

Consider $V = \mathbb{R}^{2n}$ with the Euclidean inner product and assume that

$$\omega(z, w) = \langle z, Aw \rangle .$$

Then A is nondegenerate and antisymmetric: $A^\top = -A$. Thus $iA \in \operatorname{Mat}_{2n}(\mathbb{C})$ is Hermitian, which means

its eigenvalues are all real, so A's eigenvalues are all pure imaginary. Furthermore the eigenvalues come in pairs $\pm i\alpha_j$ for $j = 1, \dots, n$, with $\alpha_j > 0$.

As iA is Hermitian, we can find $z_j = u_j + iv_j \in \mathbb{C}^{2n}$ an orthonormal basis of eigenvectors:

$$Az_j = i\alpha_j z_j, \qquad \overline{z_j}^\top z_k = \delta_{jk}.$$

And $\overline{z}_j$ fill out the basis of eigenvectors, so we have that $z_j^\top z_k = 0$. Thus

$$Au_j = -\alpha_j v_j, \qquad Av_j = \alpha_j u_j, \qquad u_j^\top v_k = u_j^\top u_k = v_j^\top v_k = 0 \text{ for } j \neq k.$$

And so

$$\omega(u_j, v_j) = u_j^\top A v_j = \alpha_j |u_j|^2 > 0,$$

and similarly $\omega(u_j, v_k) = \omega(u_j, u_k) = \omega(v_j, v_k) = 0$ for $j \neq k$. We can thus rescale u_j, v_k to get the desired basis.

Let $r = (r_1, \dots, r_n)$ be an n-tuple with $0 < r_1 \leq \cdots \leq r_n$, and define the ellipsoid

$$E(r) = \left\{ z \in \mathbb{C}^n \;\middle|\; \sum_{i=1}^n \left| \frac{z_i}{r_i} \right|^2 \leq 1 \right\}.$$

Lemma 1.4.6

Any ellipsoid

$$E = \left\{ w \in \mathbb{R}^{2n} \;\middle|\; w^\top A w \leq 1 \right\}$$

is symplectomorphic to a unique $E(r)$.

Proof

We can write E as $\left\{ w \in \mathbb{R}^{2n} \;\middle|\; g(w, w) \leq 1 \right\}$ for the inner product

$$g(u, v) = u^\top A v.$$

By the above lemma there is a basis $u_1, \dots, u_n, v_1, \dots, v_n$ of $\mathbb{R}^{2n}$ which is both symplectic and orthogonal for g. Define r_j so that $g(u_j, u_j) = g(v_j, v_j) = 1/r_j^2$. We may rearrange the u_j, v_j so that the r_j are ascending; this does not affect them forming a symplectic basis.

Let $\Phi \colon \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ be the symplectomorphism mapping the standard basis to this new basis:

$$\Phi(z) = \sum_{j=1}^n (x_j u_j + y_j v_j), \qquad z = (x_1, \dots, x_n, y_1, \dots, y_n).$$

Thus

$$g(\Psi z, \Psi z) = \sum_{j=1}^n \frac{x_j^2 + y_j^2}{r_j^2},$$

which means that

$$\Psi^{-1} E = E(r),$$

as desired.

Now to show uniqueness of $r = (r_1, \dots, r_n)$, define the matrix

$$\Delta(r) = \operatorname{diag}(1/r_1^2, \dots, 1/r_n^2, 1/r_1^2, \dots, 1/r_n^2)$$

so that

$$E(r) = \left\{ z \in \mathbb{R}^{2n} \;\middle|\; z^\top \Delta(r) z \leq 1 \right\}.$$

Because $\Psi E(r)$ maps to the ellipsoid defined by $\Psi^\top \Delta(r)\Psi$, we want to show that if $\Psi^\top \Delta(r)\Psi = \Delta(r')$, then $r = r'$. We know that $J_0\Psi^\top = \Psi^{-1}J_0$ so

$$\Psi^{-1}J_0\Delta(r)\Psi = J_0\Delta(r').$$

And so $J_0\Delta(r)$ and $J_0\Delta(r')$ have the same eigenvalues. But these matrices correspond to the complex matrices $i\,\mathrm{diag}(1/r_1^2, \ldots, 1/r_n^2)$ and $i\,\mathrm{diag}(1/{r'_1}^2, \ldots, 1/{r'_n}^2)$. Thus we must have $r = r'$ as desired. ■

Note that $E(r, \ldots, r)$ is just $B^{2n}(r)$.

Note:
In the case of $n = 1$, symplectic forms are volume forms and symplectomorphisms are the same as area-preserving maps. Thus the above lemma states that every ellipse in $\mathbb{R}^2$ can be mapped into a circle by an area-preserving linear transformation, and that this circle is unique.

Define the **symplectic spectrum** of an ellipsoid E to be the unique n-tuple $r = (r_1, \ldots, r_n)$ with $0 < r_1 \leq \cdots \leq r_n$ such that E is symplectomorphic to $E(r)$. This is an invariant which uniquely determines the ellipsoid up to symplectomorphism. So if E has symplectic spectrum r, we see that

$$\mathrm{vol}\, E = \mathrm{vol}\, E(r) = \frac{\pi^n}{n!}\prod_{j=1}^{n} r_j^2$$

where the last equality is just an explicit integral.

Theorem 1.4.7

Let $E \subseteq \mathbb{R}^{2n}$ be an ellipsoid, then

$$w_L(E) = \sup_{B\subseteq E} w_L(B) = \inf_{Z\supseteq E} w_L(E),$$

where the supremum runs over all linear (or affine) symplectic balls contained in E, and the infimum over all linear symplectic cylinders containing E. Let $r = (r_1, \ldots, r_n)$ be E's symplectic spectrum, then

$$w_L(E) = \pi r_1^2.$$

Proof

Suppose $\Psi \in \mathrm{Sp}(2n)$ is a symplectomorphism such that $\Psi E = E(r)$. Then we know

$$B^{2n}(r_1) \subseteq E(r) \subseteq Z^{2n}(r_1) \Longrightarrow \Psi^{-1}B^{2n}(r_1) \subseteq E \subseteq \Psi^{-1}Z^{2n}(r_1).$$

The widths of the bounds are both πr_1^2 so we have

$$w_L(E) = \pi r_1^2.$$

And furthermore we then get

$$\inf_{Z\supseteq E} w_L(Z) \leq \pi r_1^2 \leq \sup_{B\subseteq E} w_L(B)$$

because $\Psi^{-1}Z^{2n}(r_1)$ is in the infimum on the left, and $\Psi^{-1}B^{2n}(r_1)$ is in the supremum on the right. But if $B \subseteq E$ is a symplectic ball and $E \subseteq Z$ is a symplectic cylinder, then $w_L(B) \leq w_L(Z)$ by monotonicity, and so every element in the supremum is less than every element in the infimum, so we have equality. ■

For an ellipsoid E, define its **dual** to be

$$E^* = \left\{ v \in \mathbb{R}^{2n} \;\middle|\; \langle v, e\rangle \leq 1 \text{ for all } e \in E \right\}.$$

Let us temporarily denote the ellipsoid determined by the positive definite quadratic form A by $E(A)$, and the inner product $g_A(v,u) = v^\top Au$. Then we claim $E(A)^* = E(A^{-1})$. Notice that $v \in E(A^{-1})$ iff $v^\top A^{-1} v \leq 1$ iff $A^{-1}v \in E(A)$. Thus for all $e \in E(A)$ we have by Cauchy-Schwarz

$$\langle v, e\rangle = g_A(A^{-1}v, e) \leq 1$$

so $v \in E(A)^*$. Conversely if $v \in E(A)^*$ then

$$\langle v, e\rangle = g_A(A^{-1}v, e) \leq 1$$

for all $e \in E(A)$. Let $\alpha = \sqrt{g_A(A^{-1}v, A^{-1}v)}$ be $A^{-1}v$'s norm, so that $A^{-1}v/\alpha$ has g_A-norm 1 and thus is in $E(A)$. But

$$g_A(A^{-1}v, A^{-1}v/\alpha) = \alpha \leq 1$$

meaning $A^{-1}v \in E(A)$ and thus $v \in E(A^{-1})$ as desired.

Note that A being positive definite means A^{-1} is too, so $E(A^{-1}) = E(A)^*$ is an ellipsoid. Furthermore it is immediate from this that $E^{**} = E$ and $(\Psi E)^* = (\Psi^\top)^{-1}E^*$, the latter being because

$$(\Psi E(A))^* = E((\Psi^\top)^{-1}A\Psi^{-1})^* = E(\Psi A^{-1}\Psi^\top) = (\Psi^\top)^{-1}E(A)^*.$$

1.5 Linear complex structures

Definition 1.5.1

A **(linear) complex structure** on a real vector space is an automorphism

$$J\colon V \to V \qquad \text{such that} \qquad J^2 = -I.$$

Denote the space of linear complex structures over V by

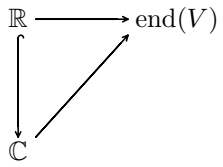
$$\mathcal{J}(V) \subseteq \operatorname{end}(V).$$

The archetypal linear complex structure is

$$J_0 = \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}$$

over $\mathbb{R}^{2n}$.

V having a complex structure is equivalent to it having the structure of a $\mathbb{C}$-vector space compatible with its real vector space structure. This is equivalent to the existence of a ring map $\mathbb{C} \to \operatorname{end}(V)$ which is a lifting of the ring map $\mathbb{R} \to \operatorname{end}(V)$ which defines V's real linear structure:



That is, the existence of a map of $\mathbb{R}$-algebras $\mathbb{C} \to \operatorname{end}(V)$. Because $\mathbb{C}$ is generated by i as a real algebra, this is equivalent to the existence of $J \in \operatorname{end}(V)$ such that $J^2 = -I$. Spelling this out, if V has a complex structure then it becomes a complex vector space with multiplication

$$\mathbb{C} \times V \to V, \qquad (a + ib, v) \mapsto av + bJv.$$

Notice that this necessitates that V has even real dimension. This can be seen because

$$\dim_{\mathbb{R}} V = \dim_{\mathbb{C}} V \cdot \dim_{\mathbb{R}} \mathbb{C} = 2 \dim_{\mathbb{C}} V$$

and because $J^2 = -I$ means that $\det J^2 = (-1)^{\dim V}$ is positive so $\dim V$ is even.

Proposition 1.5.2

Let V be a $2n$-dimensional real vector space with complex structure J. Then there is a vector space isomorphism $\Phi\colon \mathbb{R}^{2n} \to V$ such that $J\Phi = \Phi J_0$.

Proof

Viewing V as a complex vector space, there is a basis $v_1, \ldots, v_n$. Then $v_1, \ldots, v_n, Jv_1, \ldots, Jv_n$ is a basis of the real vector space V. Mapping the standard basis $e_1, \ldots, e_n, f_1, \ldots, f_n$ to this basis gives the desired result. ■